

DataPipettor: Touch-Based Information Transfer Interface Using Proximity Wireless Communication

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ABSTRACT

As diverse computing devices, including wearable and embedded systems, become ubiquitous in our living spaces, sharing and transferring data between these computers has become a usual operation. This paper introduces DataPipettor, an information transfer interface that enables intuitive data movement based on touch interactions. DataPipettor utilizes proximity wireless communication (PWC), allowing high-speed real-time communication between proximate channels while simultaneously achieving touch sensing and data transmission/reception. The interface can be miniaturized due to the characteristics of PWC, making it suitable for use in wearable devices and small equipment interfaces. Users wearing the interface can intuitively transfer data through touch interactions as if physically passing objects. This paper conducted the development and evaluation of the interface, a conceptual demonstration, and discuss future prospects.

KEYWORDS

wearable computing, touch interaction, proximity wireless communication

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1 INTRODUCTION

As Weiser said [10], technological advances have made computing devices smaller, more diverse, and affordable, leading to their ubiquity in our lives. In addition, there has been a vision of embedding tiny computer chips into everything in daily life [5]. This has increased the need for easier and more convenient inter-device information sharing. However, existing data transfer methods, like cloud services or synchronization using Wi-Fi/Bluetooth, often require technical literacy or complex setups and pairing, even for nearby

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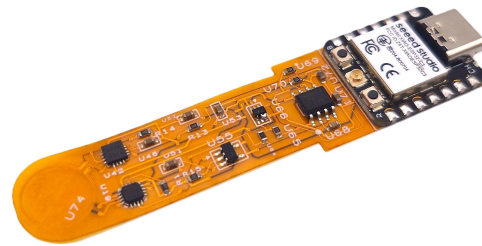


Figure 1: The overview of DataPipettor

devices. This paper proposes a touch-based information transfer interface using inductive coupling communication. It offers intuitive, high-speed (170Mb/s) data transfer and contact-based pairing, eliminating the need for complex configurations and providing a more natural user experience.

2 INTUITIVE INFORMATION TRANSFER

A seminal work in intuitive information transfer field is Pick-and-Drop [9]. This research provides an intuitive user experience by assigning the metaphor of physically picking up and moving objects to data displayed on a screen and a stylus pen. Building on this concept, subsequent studies have explored various approaches based on human gestures [3, 8, 12] or enabling seamless data transfer within specific contexts [2, 11]. Most of these research presuppose established pairing under wireless technologies such as Bluetooth or WiFi for actual data transfer. For instance, using NFC transmit small keys, and then the data is sent using faster methods. However, in dynamic environments with multiple device options, users are often compelled to re-pair devices or connect to networks before interaction, relying on non-intuitive GUI-based operations. And also as devices proliferate, broadcast methods may face bandwidth limitations. Unlike NFC, our interface has practical transfer speed, without relying on other communication technologies.

3 PROPOSED INTERFACE

We proposed DataPipettor, an interface that utilizes PWC, to facilitate information transfer based on contact and touch interactions. PWC is a technology primarily researched for high-bandwidth connections between semiconductor chips [1, 6, 7]. By eschewing broadcast communication and establishing connections only at close proximity, we achieve an intuitive pairing experience where

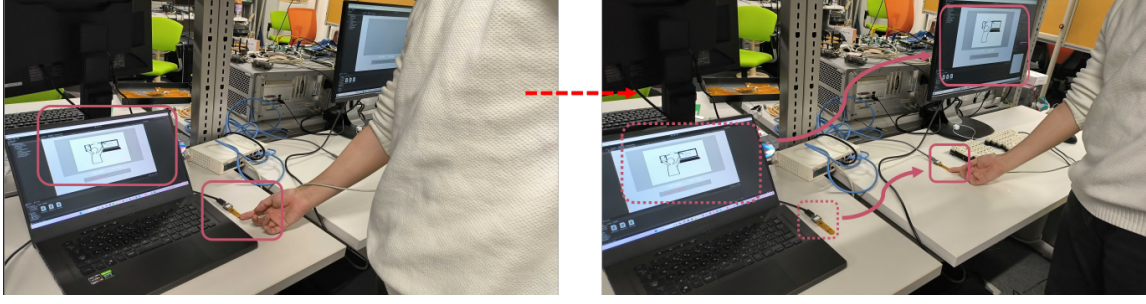


Figure 2: Using a DataPipettor to transfer data between two computers.

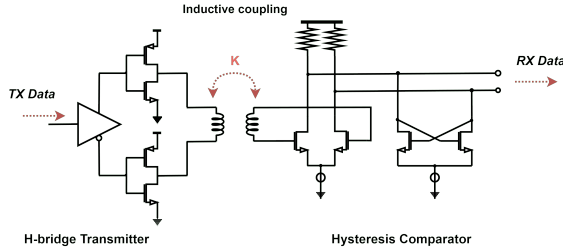


Figure 3: Transceiver core circuit

physical contact initiates communication and separation terminates it. In this paper, we developed a wearable device equipped with an interface that can be attached to users (Figure 1). This interface serves both as a wearable device for users and as a communication interface for devices. Users can transfer data through contact with the tip coil.

4 IMPLEMENTATION AND EVALUATION

PWC leverages inductive coupling, allowing data transfer without signal modulation. This characteristic enables the use of low Q -values, avoiding peaking effects. Consequently, the transceiver circuits can be implemented with high-resistance, low-turn coils, resulting in a thin, compact design. Figure 3 shows CMOS-implemented transceiver circuit. The transmitter comprises an H-bridge circuit constructed from combined inverters, while the receiver primarily employs a hysteresis comparator. Communication follows this sequence, non-return-to-zero (NRZ) baseband signal is input to the transmitter, generating a corresponding current direction in the transmitter’s coil. As the data transitions, the current direction changes, inducing a pulse voltage in the receiver’s coil. This induced pulse voltage is then restored to the original NRZ signal by the hysteresis comparator. For this study, we implemented a prototype using discrete components based on the mentioned circuit. ESP32-C3 is used as the processing unit and it has 4MB flash memory allowing to store data when used in wearable devices. The necessary parameters were experimentally determined, drawing from existing research [4], and constrained by the interface’s placement place the coil on the nail and the sufficient size (10-mm diameter) for unimpeded touch-based communication. In this prototype, it operates with a power supply of 5V and 30mA, and the inductors is directly etched into the PCB. And the experiments of this paper

,the prototype was supplied power by connecting a cable ; however, this issue can be resolved by integrating power into devices.

Experiment evaluation was conducted focused on the communication component (coil) performance, and system assessment and proof-of-concept using the entire system. For the performance evaluation, we connected the transmitting and receiving coils to a bit error rate tester (BERT) and measured the BER using Pseudo Random Binary Sequences. While a BER below 10^{-8} is generally considered the standard for practical communication lines, we defined successful communication as achieving a BER below 10^{-10} . We evaluated the maximum speed and reliability, with the latter defined as the significant proportion within one period of the digital signal (1UI). The results showed that with a 10mm coil, the maximum communication speed was 170 Mb/s. And a reliability of 0.985UI, under conditions of a 2^{7-1} PRBS signal at 150 Mb/s, indicating a sufficiently wide unit interval. Additionally, we investigated the impact of coil misalignment. The results showed that communication fails when the misalignment of the center axis of the circle exceeds approximately half of the radius. To evaluate the system, we developed an application and conducted experiments on data transfer, including images and text, between two devices mediated by a user (Figure 2). The results demonstrated that pairing and communication could be initiated through physical contact with the interface, enabling data transfer in a manner analogous to physically moving objects. The system successfully transferred data sizes of up to approximately 10 KBytes, confirming its practical functionality for common file types.

5 CONCLUSION

In this paper, we introduced DataPipettor, a compact communication interface utilizing PWC. This interface offers intuitive data transfer, due to sufficient transfer speeds and pairing through physical contact. With this interface, users can intuitively transfer data without the complexities of network environments or pairing procedures. There are the issue of communication interruption when coil alignment is not precise during use, and the fact that the overall system has not yet achieved sufficient communication speeds due to factors such as parasitic capacitance. As a future work, we intend to conduct detailed parameter analysis and arrange the smaller coils in an array and realize multichannel to enforce the robustness and communication speed. Furthermore, we aim to miniaturize through integrated circuit design, also incorporating power supply and memory.

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